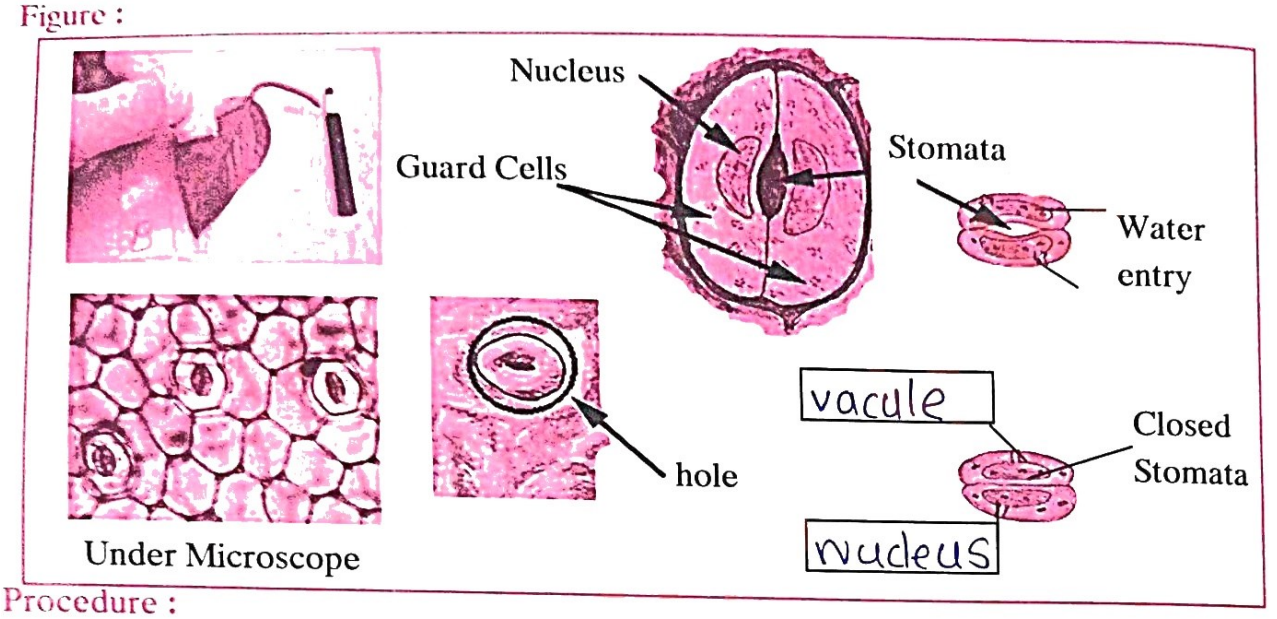
Experiment No.9

Title : To study stomata by making temporary slide of leaf peel.

Materials : Leaf of any dicot plant, Compound Microscope, Slides, Coverslips, Dropper,

Chemicals : Iodine and Glycerine.

Figure :



Procedure :

1. Take a leaf of any dicot plant and observe various parts of it. In case of small leaf, place it on glass slides and put a drop of glycerine on it. Cover it with coverslip and observe under microscope.
2. To study stomata select any potted dicot plant and keep it few hours in sunlight.
3. Remove one leaf from the above potted plant and tear it tangentially to get thin peel of lower side of leaf.
4. Take a small piece of peel on the slide and put a drop of dilute iodine solution.
5. Keep the coverslipon it and keep the slide in the sunlight for a minute.
6. Observe the slide under the compound microscope and note down the observations.
7. Keep the same slide in the classroom for few minutes and now observed it again under under the microscope. Note down the observation.

Observation :

A number of small all epithical pores a caused to the a Stomatal structure is different in both dicot and monocot leaves gaurd cells of stomata.

Inference :

Stomata in sunlight : stomata are exhaled water in the form of vapour.

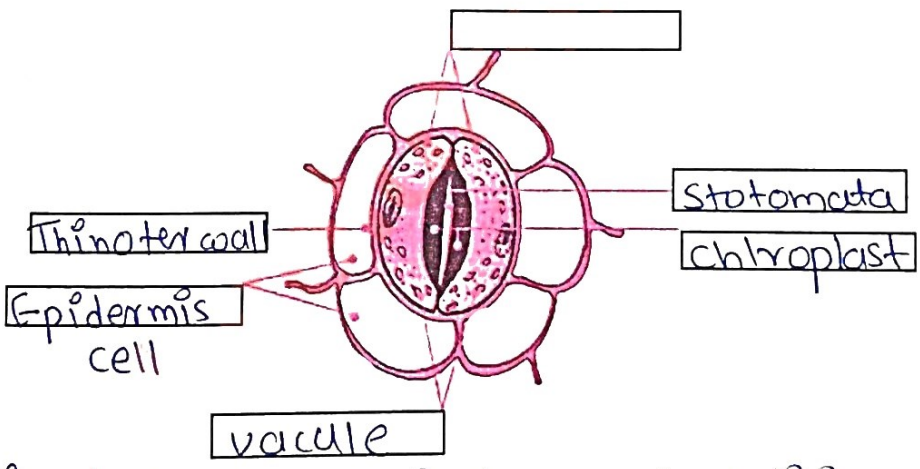
Stomata in dark/ absence of sunlight : stomata are closed in the dark in moist plants.

Multiple Choice Qusetions

1. To see the stomata in the plant leaves .....is used to make a slide.
   A. crushed leaf
   B. stalk of leaf/ upper epidermis
   C. lower surface of a leaf
   D. central part of leaf
2. When lower peel of the leaf is observed under the compound microscope, the colour of was redish pink due to.....stain.
   A. Iodine
   B. Acetocarmine
   C. Safranin
   D. Eosin
3. While preparing a temporary stained mount of leaf epidermal peel, the extra stain is removed by .....
   A. washing with water
   B. washing with calcium chloride
   C. soaking with filter paper
   D. absorbing with cotton wool
4. Leaf of monocot and dicot plant can be distinguished on the basis of .....
   A. Shape of stomata
   B. thickness of epidermis
   C. size of cells
   D. position of stomata

Questions

1. Label the following diagram and write the explanation about it.



conclusing 1) stomata are minute apprature living on the upper air lower surface of the leaf they are guarded by a pair of specical lizard ~~dtd~~ ediclemal cell called guard cell.

2. Why the number of stomatas present on upper surface of a leaf are comparetively less than on the lower surface of leaf?

The number of stomata present on upper surface of a leaf are comparatively less to present excessive water loss through as it faces direct sunlight.

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